

# Mason J. Fleming Biomedical Engineering Student & Researcher

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## EDUCATION

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**Virginia Polytechnic Institute and State University** , 08/2026 – 05/2030  
*Doctor of Philosophy in Biomedical Engineering* Blacksburg, VA  
Overall GPA: N/A

**West Virginia University** , 08/2021 – 05/2026  
*Bachelor of Science in Biomedical Engineering (ABET-accredited)* Morgantown, WV  
Minor in Computer Science  
Overall GPA: 3.76

## RESEARCH AND PROFESSIONAL EXPERIENCE

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**Undergraduate Researcher, West Virginia University (Statler College)** , 09/2024 – Present  
Morgantown, WV

- Developed and implemented a force sensor system for medication adherence monitoring in an NIH-funded study, focusing on sensor integration, signal processing, and data analysis under Dr. Stephen Cain and Ph.D. candidate Daniel Duque Urrego.
- Assisted with implementation and execution of an updated BMEG 420L, by preparing equipment and performing tests on the lab's projects and assignments.
- Participated in the WVU Summer Undergraduate Research Program (SURE) during Summer 2025.

**Student Grader (BMEG 201 & BMEG 230), West Virginia University (Statler College)** , 09/2024 – Present  
Morgantown, WV

- Worked under Dr. Anika Pirkey as a grader for two different biomedical engineering classes, Introduction to Biomedical Engineering (BMEG 201) and Numerical Methods in Biomedical Engineering (BMEG 230).
- Graded homework assignments, quizzes, and exams completed by the students in a timely manner and in collaboration with other graders and Dr. Anika Pirkey.

**Outreach & Recruitment Office Staff, West Virginia University (Statler College)** , 06/2024 – Present  
Morgantown, WV

- Supported Statler College outreach and recruitment by leading campus tours, facilitating STEM activities for K-12 students, and representing the college at full-day university outreach events.
- Participated in multiple tabling events, college fairs, and Q&A sessions geared toward undergraduate recruitment and K-12 outreach.

**Eugene V. Cilento Learning Center Tutor, West Virginia University (Statler College)** , 09/2023 – Present  
Morgantown, WV

- Provided academic support to students in introductory engineering, science, and mathematics courses, focusing on foundational technical concepts and problem-solving.

**Engineering Challenge Camp Counselor, West Virginia University (Statler College)** , 05/2023 – 07/2023  
Morgantown, WV

- Facilitated and taught multiple week-long day camps focused on STEM and engineering activities and projects to K-12 students.

## ABSTRACTS & PRESENTATIONS

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| <b>Van Liere and WV IDeA Research Conference</b> ☑, <i>West Virginia University</i><br><b>Fleming, M.J.</b> , Inamdar, K., Gritsenko, V., Miller, S., Copenhaver, I., Whitlock, R.<br><i>"De'clinicalizing' Infant Motor Assessment: How Might AI Transform Accessibility and Ecological Validity?"</i> | 10/2025<br>Morgantown, WV |
| <b>18th Annual Summer Undergraduate Research Symposium</b> , <i>West Virginia University</i><br><b>Fleming, M.J.</b> , Duque Urrego, D., Cain, S. <i>"Continued Development: Prototype Force Sensing Bottle to Accurately Measure Self-Administration of Eyedrop Medication"</i>                        | 07/2025<br>Morgantown, WV |
| <b>9th Annual Spring Undergraduate Research Symposium</b> ☑, <i>West Virginia University</i><br><b>Fleming, M.J.</b> , Duque Urrego, D., Cain, S. <i>"Development of a Prototype Force Sensing Bottle to Accurately Measure Self-Administration of Eyedrop Medication"</i>                              | 04/2025<br>Morgantown, WV |
| <b>Biomedical Engineering Capstone Poster Symposium</b> , <i>West Virginia University</i><br><b>Fleming, M.J.</b> , Inamdar, K., Gritsenko, V., Copenhaver, I., Whitlock, R., Miller, S.,<br><i>"Validation and Use of Markerless Motion Capture to perform Infant Movement Assessments"</i>            | 04/2026<br>Morgantown, WV |

## PROJECT AND CLASS EXPERIENCE

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| <b>Biomedical Engineering Senior Design (BMEG 455/BMEG 456)</b> , <i>Capstone Project</i><br><i>Chief Engineer   Sensor and AI-Driven Smart Tech for Infant Movement Analysis</i> <ul style="list-style-type: none"><li>Under the mentorship of Dr. Anika Pirkey and Ph.D. candidate Spencer Miller, worked directly with Dr. Ketaki Inamdar and Dr. Valeriya Gritsenko to further develop techniques and technology to diagnose and measure atypical neural development in infants 6 months of age or younger.</li><li>Developed and processed markerless motion capture data from videos of various infants from in the lab using MATLAB and Python.</li></ul> | 08/2025 – 05/2026 |
| <b>Biomechanics, West Virginia University (BMEG 340)</b> ,<br><i>Impact of Weighted Vest on Gait Mechanics</i> <ul style="list-style-type: none"><li>Performed 3D Motion Capture on an instrumented treadmill to analyze the effect of a weighted vest on joint angles and forces.</li></ul>                                                                                                                                                                                                                                                                                                                                                                     | 01/2025 – 05/2025 |
| <b>Biomedical Imaging (BMEG 310)</b> ,<br><i>Analysis of SPECT/MRI Imaging for Early Diagnosis of Parkinson's Disease</i> <ul style="list-style-type: none"><li>Determined the value of SPECT and MRI in diagnosis of Parkinson's Disease, and what the various imaging modalities can reveal.</li></ul>                                                                                                                                                                                                                                                                                                                                                         | 01/2025 – 05/2025 |
| <b>Biomaterials, West Virginia University (BMEG 311)</b> , <i>Analysis and Proposed Design Modifications to the Vivistim Paired VNS System by MicroTransponder Inc.</i> <ul style="list-style-type: none"><li>Analyzed the efficacy of the Vivistim system for stroke rehabilitation. Proposed two design modifications to modify the size, feel, and communication abilities of the Vivistim system.</li></ul>                                                                                                                                                                                                                                                  | 08/2024 – 12/2024 |
| <b>Biomedical Instrumentation (BMEG 420)</b> ,<br><i>Evaluation of the Physiological Response to Spice Levels</i> <ul style="list-style-type: none"><li>Instrumented galvanic skin response (GSR) and electrocardiogram (ECG) sensors in order to measure the effect different name brand hot sauces had on heart and perspiration rate.</li></ul>                                                                                                                                                                                                                                                                                                               | 08/2024 – 12/2024 |

**Introduction to Biomedical Engineering, West Virginia University (BMEG 201),**

08/2023 – 12/2023

*Powered Toe Prosthetic Design and Analysis*

- Designed, developed, and mathematically analyzed an above the knee prosthetic with a powered knee and actuating toes for stability and balance.

**Germany: Explore German Technology and Culture, Munich, Germany (ENGR 230),**

03/2023

*Spring Break Study Abroad Experience – West Virginia University*

- Visited Rehau Group factories, MAN Truck and Bus factory, and the Zugspitze cable car, as well as many cultural sites over ten days as part of a university-sponsored spring break study abroad trip.

**TECHNICAL SKILLS**

**Programming Languages**

● ● ● ● ●

MATLAB, Arduino, C, CSS, Java, JavaScript

**Circuit Design**

● ● ● ● ●

LTSpice, Fusion360

**CAD**

● ● ● ● ●

SolidWorks, Fusion360

**CPR/First Aid**

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American Red Cross Certification

**EXTRACURRICULAR ACTIVITIES**

**West Virginia University Boxing Club, *President***

01/2024 – Present

Oversee all operations of the student-run boxing club, compliant with USA Boxing and the National Collegiate Boxing Association organizations. Helped raise funds, grow the club, host boxing events and connect with the Morgantown community by participating in community service, all whilst competing in the aforementioned organizations' official bouts, matches, and tournaments.

Morgantown, WV, United States

**Benjamin M. Statler College of Engineering and Mineral Resources [↗](#), *Ambassador***

08/2023 – Present

Serve as an exemplary undergraduate student within the Benjamin M. Statler College of Engineering and Mineral Resources, maintaining an engaged and positive role within the college and Department of Chemical and Biomedical Engineering.

Morgantown, WV, United States

**Biomedical Engineering Society [↗](#), *Member***

08/2021 – Present

Active member in the West Virginia University student chapter of the Biomedical Engineering Society (BMES). Attend regular monthly meetings, stay informed on the current state of biomedical engineering in West Virginia, the United States, and abroad.

Morgantown, WV, United States